Abstract

Modern educational institutions require secure, scalable, and well-structured network infrastructures to support academic, administrative, and student services efficiently. Traditional flat networks often suffer from broadcast congestion, limited security, and poor scalability, especially when multiple departments and remote locations are involved. This project focuses on the design, implementation, and testing of a hierarchical campus network for a multi-site educational institution using Cisco Packet Tracer.

The proposed network architecture connects three geographically separated locations: the Main Campus, the Academic Block, and the Student Services Center. At the Main Campus, Virtual Local Area Networks (VLANs) are implemented to logically segment departments including Administration, Faculty, IT Support, Library, and Accounts, thereby improving performance, security, and manageability. Inter-VLAN communication is enabled using a Router-on-a-Stick approach, allowing controlled communication between departments. The remote sites are designed with a simplified structure using a single VLAN per site to ensure ease of management while maintaining connectivity with the main campus.

A scalable IP addressing scheme based on subnetting is applied across the entire network to efficiently utilize address space and support future expansion. Wide Area Network (WAN) connectivity is established between all sites using point-to-point serial links. To ensure dynamic and reliable inter-site communication, the Open Shortest Path First (OSPF) routing protocol is implemented on all routers. Additionally, Access Control Lists (ACLs) are configured to enforce security policies by restricting unauthorized access to sensitive resources such as the Accounts department and centralized servers.

INTRODUCTION

In today's digital era, educational institutions heavily rely on computer networks to support academic, administrative, and operational activities. As institutions expand by adding new buildings and departments, network complexity increases, requiring structured, secure, and scalable network architecture. A poorly designed flat network can lead to excessive broadcast traffic, security vulnerabilities, and difficulty in management and expansion.

This project addresses these challenges by designing a hierarchical campus network for a growing educational institution with three locations: the Main Campus, the Academic Block, and the Student Services Center. The primary goal is to replace the existing flat network with a segmented and secure infrastructure that ensures efficient communication, departmental isolation, and reliable connectivity across all sites.

Cisco Packet Tracer is used as the simulation tool to design and implement the network. The project incorporates key networking concepts such as VLAN segmentation, IP subnetting, inter-VLAN routing, WAN connectivity, dynamic routing protocols, and Access Control Lists (ACLs). VLANs are used to logically separate departments at the Main Campus, reducing broadcast traffic and improving security. A centralized server is deployed to provide network services such as DNS, DHCP, and web services to all campuses.

Wide Area Network (WAN) links connect the remote campuses to the Main Campus, enabling inter-site communication. Dynamic routing using the OSPF protocol ensures automatic route discovery and network adaptability in case of future expansion. Additionally, security mechanisms are implemented to protect sensitive departments and resources from unauthorized access.

Project Objectives

General Objective

The primary objective of this project is to design, configure, and implement a secure, scalable, and efficient multi-campus network infrastructure using Cisco Packet Tracer. The network is designed to support reliable communication between different departments and geographically separated campuses while ensuring proper security and controlled access to sensitive resources.

Specific Objective

The specific objectives of this project are:

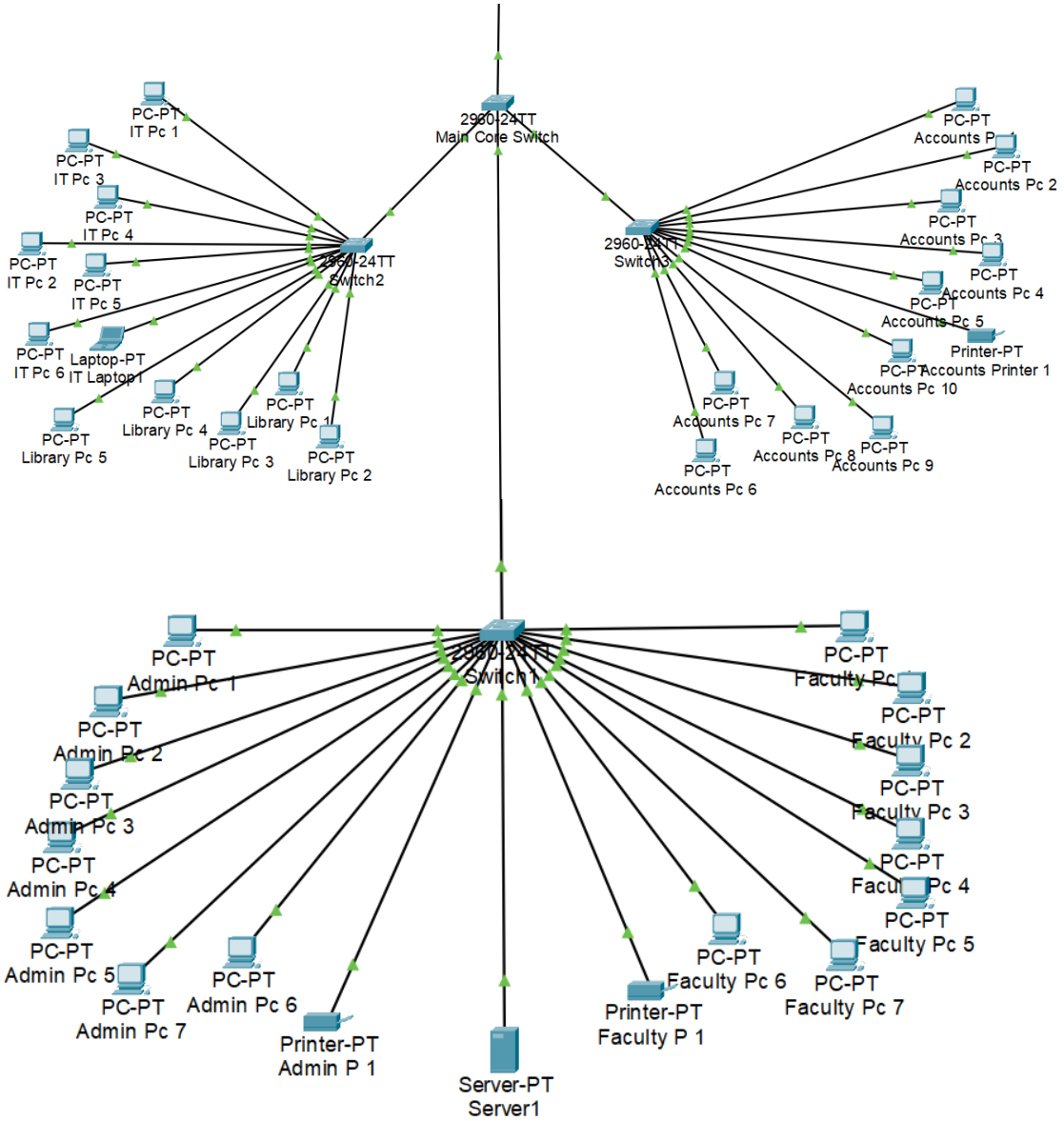
- To design a hierarchical campus network connecting the Main Campus, Academic Block, and Student Services Center.
- To implement VLAN-based departmental segmentation at the Main Campus to reduce broadcast traffic and enhance security.
- To configure inter-VLAN routing using Router-on-a-Stick methodology.
- To design and apply a scalable IP addressing scheme using subnetting.
- To establish WAN connectivity between campuses using serial links.
- To implement a dynamic routing protocol (OSPF) for inter-site communication.
- To configure Access Control Lists (ACLs) to protect sensitive VLANs and servers.
- To deploy and configure essential network services such as DNS, DHCP, and Web Server.
- To test, troubleshoot, and verify the network for connectivity, performance, and security.

Network Design Overview

Campus Network Architecture

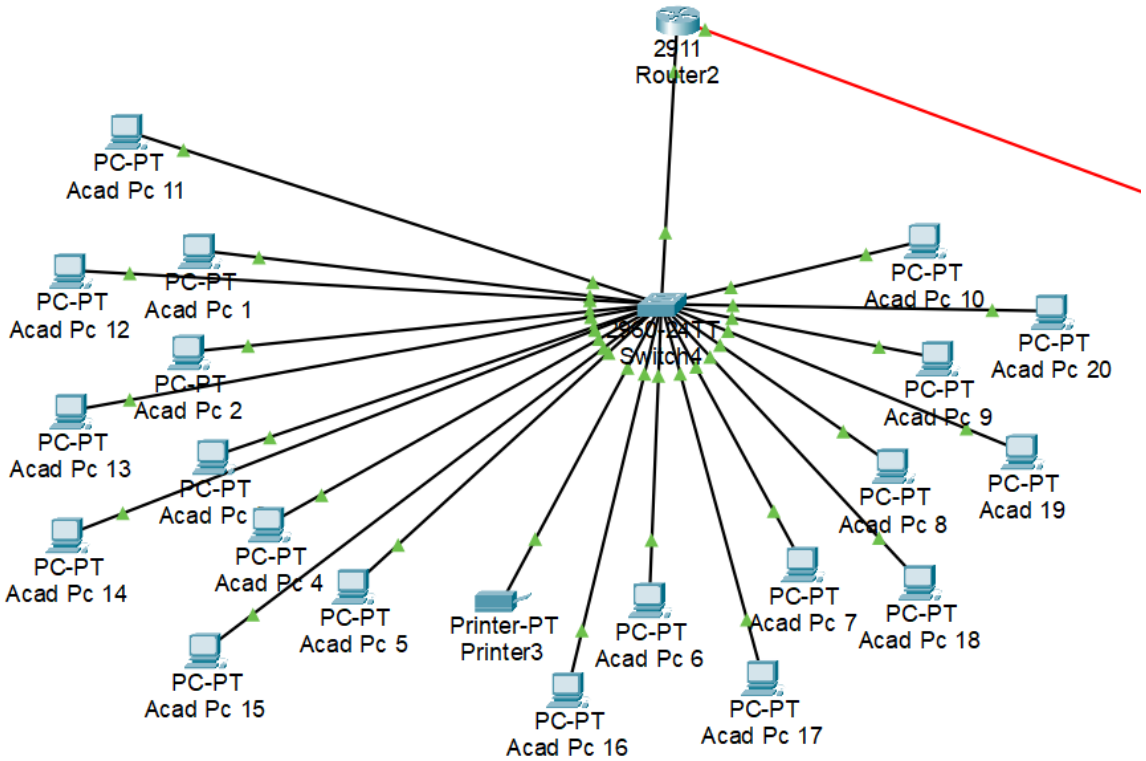
The proposed network follows a hierarchical network design model, which improves scalability, performance, and manageability. The architecture consists of three major locations:

Main Campus – Central site containing multiple departments and network services. At the Main Campus, multiple switches were deployed to accommodate different departments. VLANs were



used to logically separate departmental traffic while using the same physical switching infrastructure. A single router was connected to the Main Campus switches using trunk links to enable inter-VLAN routing. This design reduces broadcast traffic and improves both performance and security.

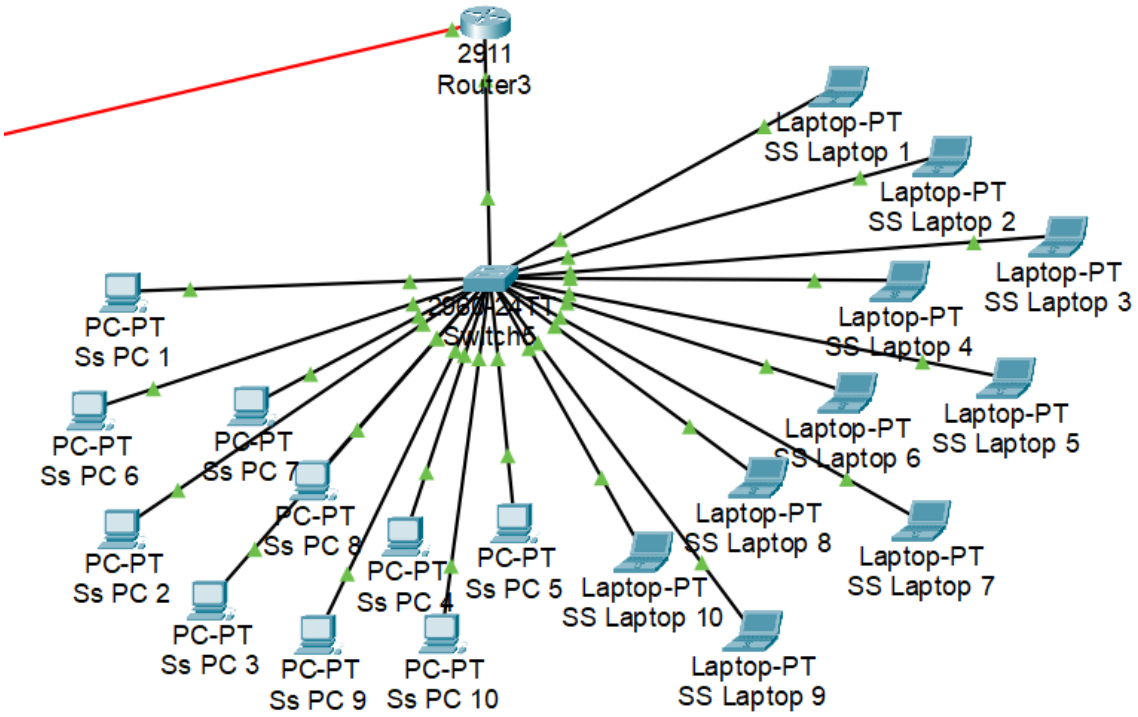
Academic Block – The Academic Block was designed as a remote campus connected to the Main Campus through a Wide Area Network (WAN). In accordance with the project documentation, this site follows a simple and cost-effective network design consisting of one router, one switch, and a single VLAN. The use of a single VLAN ensures minimal configuration complexity while still allowing reliable communication with the Main Campus. All academic PCs and end devices in this block are connected to the access switch and belong to the same VLAN. The switch is connected to the local router, which acts as the default gateway for all devices in the Academic Block.



Student Services Center – The Student Services Center was also implemented as a remote site and follows the same simplified design model specified in the project requirements. This campus includes one router, one switch, and a single VLAN, making the network easy to manage while still ensuring full connectivity with the Main Campus.

All student services devices, such as PCs used for admissions, registration, or student support, are connected to the switch and assigned to the same VLAN. The switch connects to the local router, which provides Layer-3 services and serves as the default gateway for the site. The router is linked

to the Main Campus router using a point-to-point serial WAN connection, forming a star topology with the Main Campus at the center.



Server – To support essential network services and centralized resource management, a dedicated server was deployed at the Main Campus of the network. The server was placed in a separate Server VLAN (VLAN 60) to enhance security, simplify access control, and align with best practices for enterprise network design. The purpose of using a centralized server in this project is to provide core network services that are required by users across all departments and remote campuses. In this project, the server performs the following functions:

- **DHCP Service**
  Enables name resolution so that users can access network resources and the web server using domain names instead of IP addresses.
- **Web HTTP Service**
  Hosts the project's web pages, including the welcome page describing the campus network. This allows verification of application-layer connectivity from different VLANs and remote campuses.

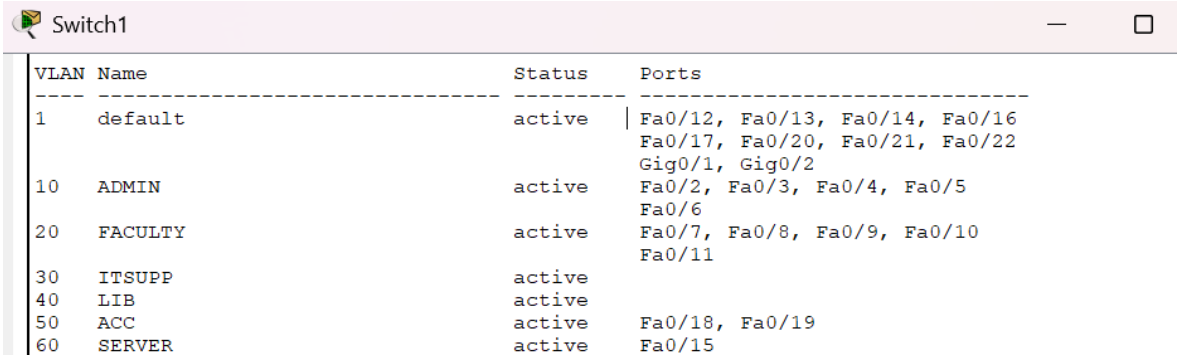
- **DHCP Service:**
  Can dynamically assign IP addresses to hosts, reducing manual configuration and supporting scalability.

VLAN Design at Main Campus

To ensure departmental isolation and reduce unnecessary broadcast traffic, Virtual Local Area Networks (VLANs) were implemented at the Main Campus.

Department	VLAN ID	Subnet
Administration	10	192.168.10.0/24
Faculty	20	192.168.20.0/24
IT Support	30	192.168.30.0/24
Library	40	192.168.40.0/24
Accounts	50	192.168.50.0/24
Servers	60	192.168.60.0/24

Each department is assigned a unique VLAN and subnet, enabling secure separation and controlled communication.



```
Switch1

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/12, Fa0/13, Fa0/14, Fa0/16
                                                Fa0/17, Fa0/20, Fa0/21, Fa0/22
                                                Gig0/1, Gig0/2
10   ADMIN                            active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6
20   FACULTY                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11
30   ITSUPP                           active
40   LIB                              active
50   ACC                              active    Fa0/18, Fa0/19
60   SERVER                           active    Fa0/15
```

Switch ports connected to departmental PCs were configured as access ports and assigned to their respective VLANs. Trunk ports were configured between switches and between the switch and the router to carry traffic for multiple VLANs using IEEE 802.1Q encapsulation. This configuration allows a single physical link to transport traffic from all VLANs while maintaining logical separation.

VLAN Design at Remote Campuses

According to the project requirements, each remote site is configured with a single VLAN:

Campus	VLAN ID	Subnet
Academic Block	70	192.168.70.0/24
Student Services Center	80	192.168.80.0/24

This simplified design ensures easy management while maintaining connectivity with the Main Campus.

- **Academic Block**

Switch4

```
VLAN Name                             Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                          active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
70   ACADEMIC                         active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10
```

- **Student Services Block**

Switch5

```
VLAN Name                             Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                          active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
80   STUDENT_SERVICE                  active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10
```

IP Addressing and Subnetting Scheme

At the Main Campus, each VLAN was allocated a /24 subnet, such as 192.168.10.0/24 for Administration and 192.168.60.0/24 for the Server VLAN. This provides sufficient IP addresses for current devices and allows room for future expansion.

The Academic Block and Student Services Center were also assigned individual /24 subnets, each corresponding to their single VLAN design.

LAN Addressing

- All LAN networks use /24 subnets for simplicity and scalability.
- Default gateways are assigned on router interfaces (e.g., `192.168.X.1`).

Router1

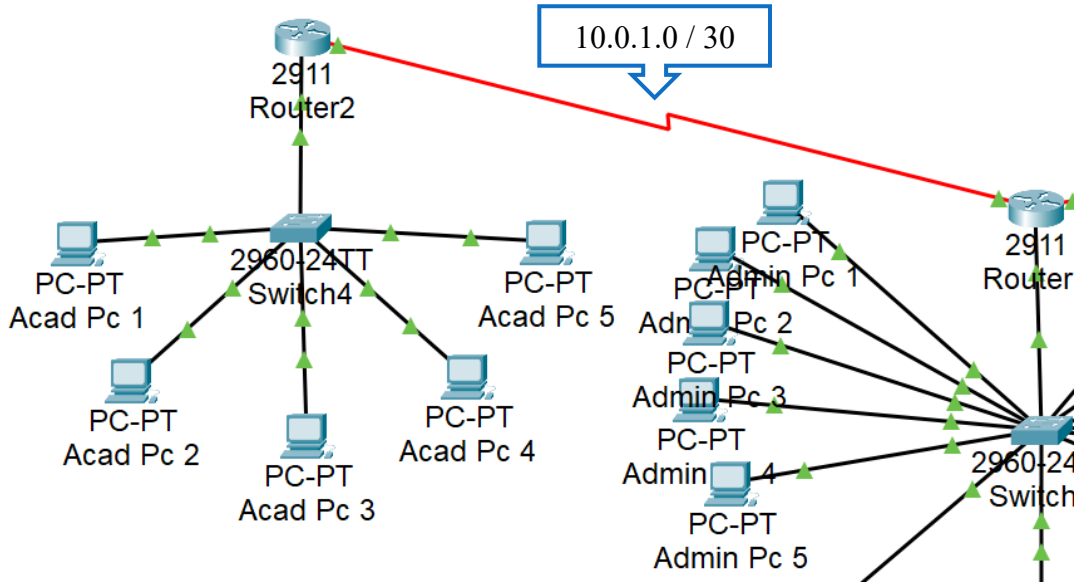
```
Router1>en
Router1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES NVRAM  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/0.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/0.40  192.168.40.1    YES manual up                    up
GigabitEthernet0/0.50  192.168.50.1    YES manual up                    up
GigabitEthernet0/0.60  192.168.60.1    YES manual up                    up
GigabitEthernet0/0.70  unassigned      YES unset  up                    up
GigabitEthernet0/0.80  unassigned      YES unset  up                    up
GigabitEthernet0/1     unassigned      YES NVRAM  administratively down down
GigabitEthernet0/2     unassigned      YES NVRAM  administratively down down
Serial0/3/0            10.0.1.1        YES manual up                    up
Serial0/3/1            10.0.2.1        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

WAN Addressing

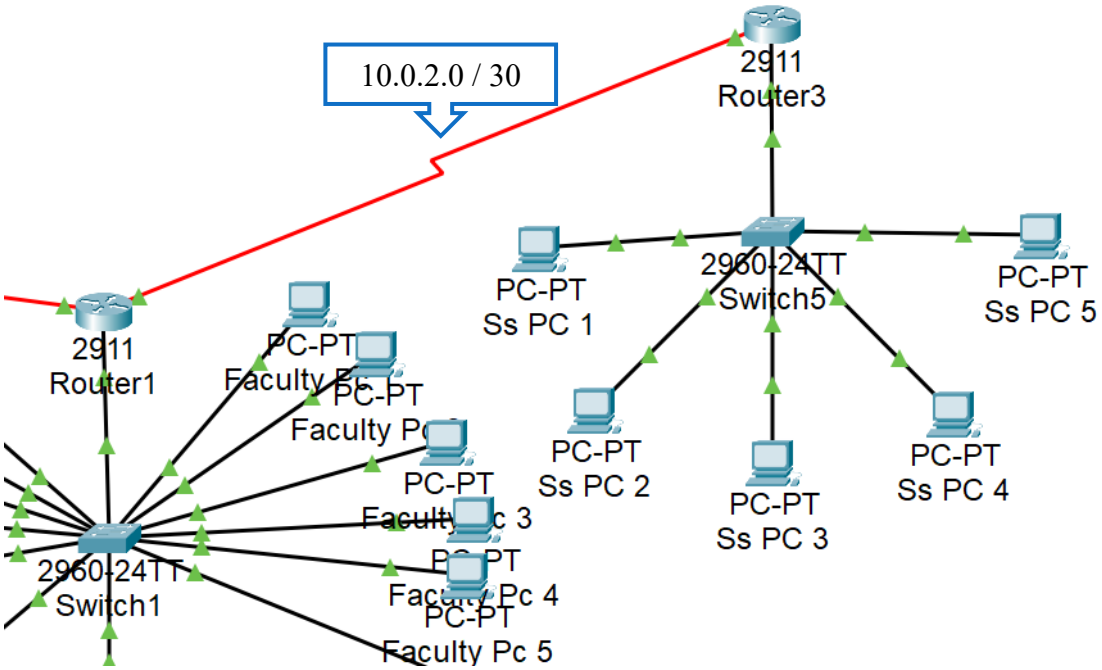
- Point-to-point WAN links use /30 subnets to minimize IP address wastage.

WAN Link	Network	Router IPs
Main ↔ Academic	10.0.1.0/30	10.0.1.1, 10.0.1.2
Main ↔ Student Services	10.0.2.0/30	10.0.2.1, 10.0.2.2

- **Main ←→ Academic**



- **Main ←→ Student Services**



Device Configuration

Switch Configuration

Switches were configured to:

- Create VLANs for all required departments.
- Assign access ports to appropriate VLANs.
- Configure trunk links between switches and routers.
- Ensure proper VLAN propagation across interconnected switches.

Router Configuration

Routers were configured to:

- Enable inter-VLAN routing using Router-on-a-Stick.
- Establish WAN connectivity using serial interfaces.
- Implement OSPF for dynamic routing.
- Act as default gateways for all VLANs and remote networks.

WAN Connectivity and Dynamic Routing

A dynamic routing protocol, OSPF (Open Shortest Path First), was implemented on all routers as required by the project documentation. OSPF allows routers to automatically exchange routing information, ensuring that all networks are reachable without the need for static routes. In this project, all LAN and WAN networks were advertised into OSPF area 0. Once configured, routers dynamically learned routes to all VLANs and remote site networks.

WAN connectivity allows remote campuses to access centralized services hosted at the Main Campus, such as DNS and Web servers, while also enabling controlled inter-campus communication. The use of dynamic routing improves reliability and scalability, making it easy to add future campuses without major reconfiguration.

- **Router 1 (Main Campus OSPF)**

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
C       192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L       192.168.40.1/32 is directly connected, GigabitEthernet0/0.40
     192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.50.0/24 is directly connected, GigabitEthernet0/0.50
L       192.168.50.1/32 is directly connected, GigabitEthernet0/0.50
     192.168.60.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.60.0/24 is directly connected, GigabitEthernet0/0.60
L       192.168.60.1/32 is directly connected, GigabitEthernet0/0.60
O    192.168.70.0/24 [110/65] via 10.0.1.2, 00:00:30, Serial0/3/0
O    192.168.80.0/24 [110/65] via 10.0.2.2, 00:00:30, Serial0/3/1
```

- **Router 2 (Academic Block OSPF)**

Router2

```
C       10.0.1.0/30 is directly connected, Serial0/3/0
L       10.0.1.2/32 is directly connected, Serial0/3/0
O       10.0.2.0/30 [110/128] via 10.0.1.1, 00:00:25, Serial0/3/0
O    192.168.10.0/24 [110/65] via 10.0.1.1, 00:00:25, Serial0/3/0
O    192.168.20.0/24 [110/65] via 10.0.1.1, 00:00:25, Serial0/3/0
O    192.168.30.0/24 [110/65] via 10.0.1.1, 00:00:25, Serial0/3/0
O    192.168.40.0/24 [110/65] via 10.0.1.1, 00:00:25, Serial0/3/0
O    192.168.50.0/24 [110/65] via 10.0.1.1, 00:00:25, Serial0/3/0
O    192.168.60.0/24 [110/65] via 10.0.1.1, 00:00:25, Serial0/3/0
     192.168.70.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.70.0/24 is directly connected, GigabitEthernet0/0
L       192.168.70.1/32 is directly connected, GigabitEthernet0/0
O    192.168.80.0/24 [110/129] via 10.0.1.1, 00:00:25, Serial0/3/0
```

- **Router 3 (Ss Block OSPF)**

Router3

```
O       10.0.1.0/30 [110/128] via 10.0.2.1, 00:00:30, Serial0/3/0
C       10.0.2.0/30 is directly connected, Serial0/3/0
L       10.0.2.2/32 is directly connected, Serial0/3/0
O    192.168.10.0/24 [110/65] via 10.0.2.1, 00:00:30, Serial0/3/0
O    192.168.20.0/24 [110/65] via 10.0.2.1, 00:00:30, Serial0/3/0
O    192.168.30.0/24 [110/65] via 10.0.2.1, 00:00:30, Serial0/3/0
O    192.168.40.0/24 [110/65] via 10.0.2.1, 00:00:30, Serial0/3/0
O    192.168.50.0/24 [110/65] via 10.0.2.1, 00:00:30, Serial0/3/0
O    192.168.60.0/24 [110/65] via 10.0.2.1, 00:00:30, Serial0/3/0
O    192.168.70.0/24 [110/129] via 10.0.2.1, 00:00:20, Serial0/3/0
```

Security Implementation Using ACLs

Network security is a critical requirement of this project, particularly for protecting sensitive departmental data and centralized services. Access Control Lists (ACLs) were used to implement controlled access policies across the network.

In this project, ACLs were specifically applied to protect the Accounts VLAN(50) and the Server VLAN(60). The Accounts department contains sensitive financial information, while the server hosts critical services such as DNS and Web applications. ACL rules were configured on the router to restrict unnecessary access to these VLANs while still allowing authorized communication.

For example, general departments were allowed limited access to the server for required services, while direct access to the Accounts VLAN was restricted. These ACLs were applied on the appropriate router sub-interfaces to ensure traffic filtering occurs at the correct point in the network.

Router1

```
Router1#show access-lists
Extended IP access list PROTECT_SENSITIVE
    10 permit ip 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255
    20 permit ip 192.168.20.0 0.0.0.255 192.168.50.0 0.0.0.255
    30 permit ip 192.168.10.0 0.0.0.255 192.168.60.0 0.0.0.255
    40 permit ip 192.168.20.0 0.0.0.255 192.168.60.0 0.0.0.255
    50 deny ip any 192.168.50.0 0.0.0.255
    60 deny ip any 192.168.60.0 0.0.0.255
    70 permit ip any any
```

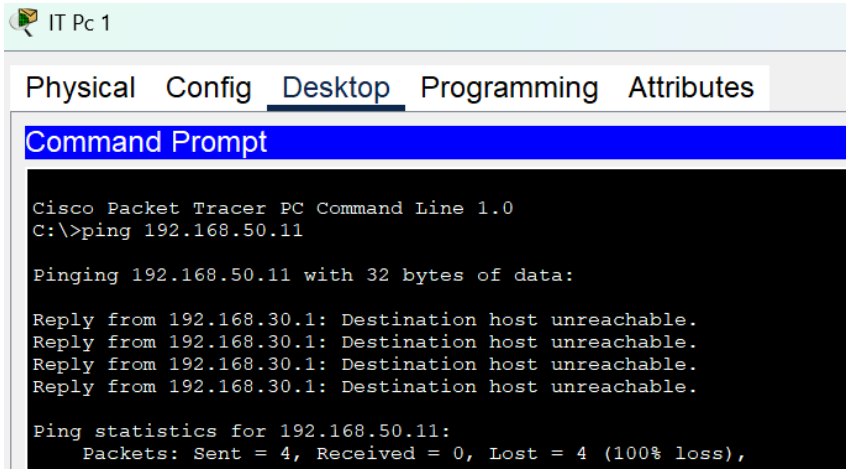
By implementing ACLs, the project demonstrates practical application of network security concepts and fulfills the requirement for controlled access policies as stated in the project documentation.

Network Verification and Validation

This section verifies that the implemented network meets all functional and security requirements defined in the project documentation. Network verification was performed after completing VLAN configuration, inter-VLAN routing, WAN connectivity, dynamic routing, and ACL implementation.

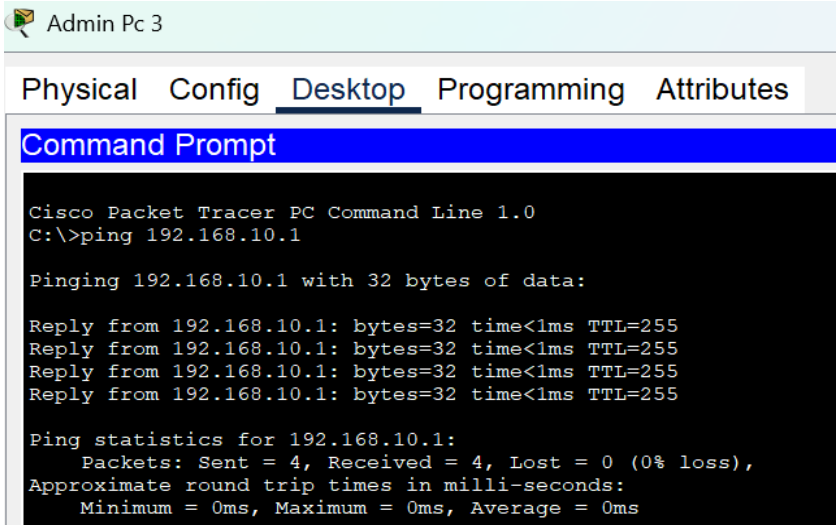
- **Inter-VLAN Connectivity Test (Main Campus)**

This test confirms that inter-VLAN routing is functioning correctly through the router-on-a-stick configuration. PCs from different VLANs at the Main Campus were able to communicate via the router.



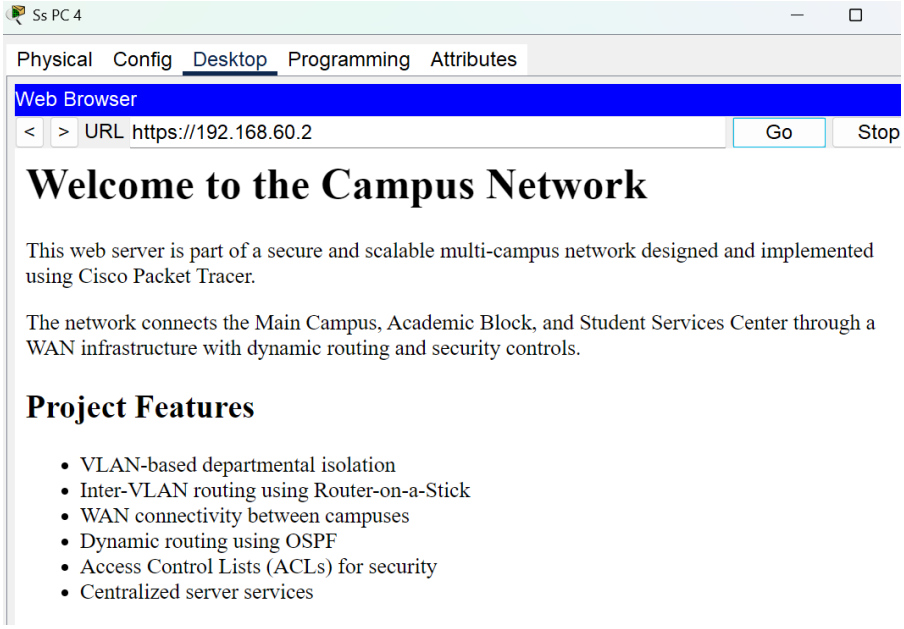
- **Default Gateway Reachability Test**

This test ensures that all PCs can reach their respective default gateways. Each PC successfully pinged its VLAN gateway IP address.



- **Server Accessibility Test (LAN)**

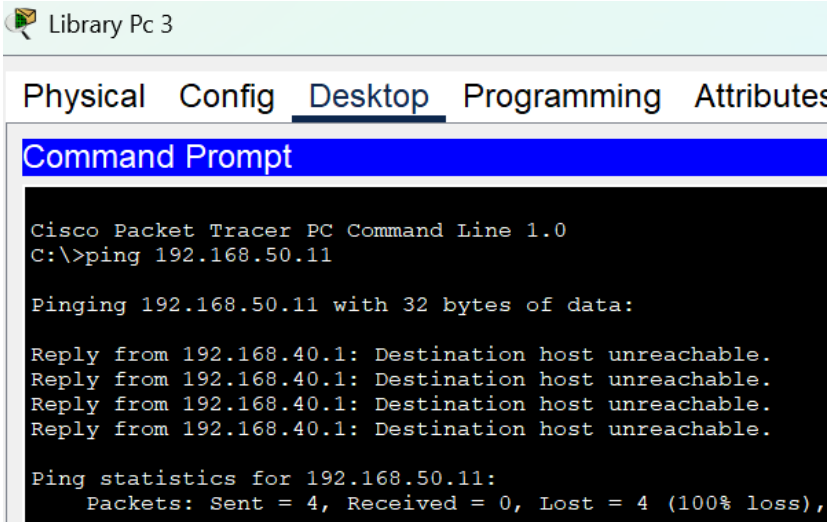
This test verifies that the centralized server at the Main Campus is reachable from permitted VLANs. Authorized VLANs successfully accessed the server services (HTTP / DNS).



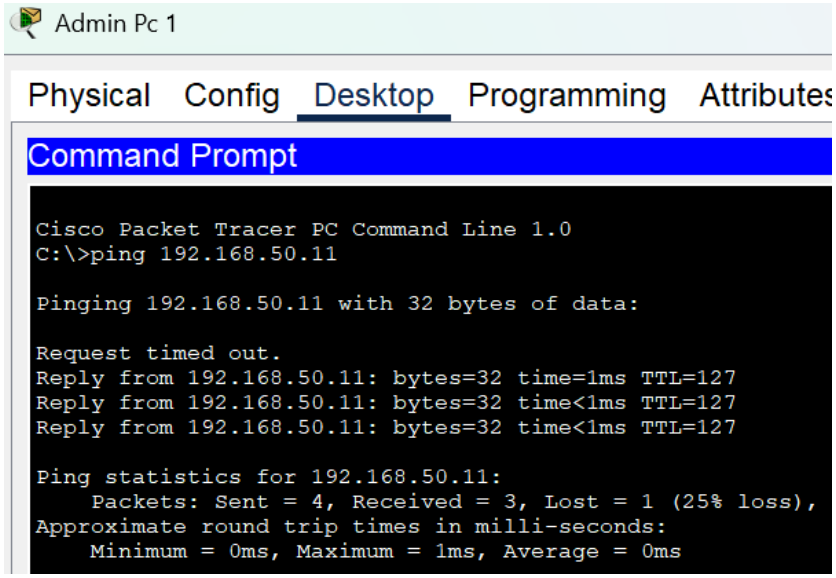
- **ACL Verification Test (Security Enforcement)**

This test confirms that ACLs are correctly restricted and permitting traffic as per security policy. Restricted VLANs were denied access to protected VLANs, while permitted traffic was allowed.

- Failed ping due to ACL restriction

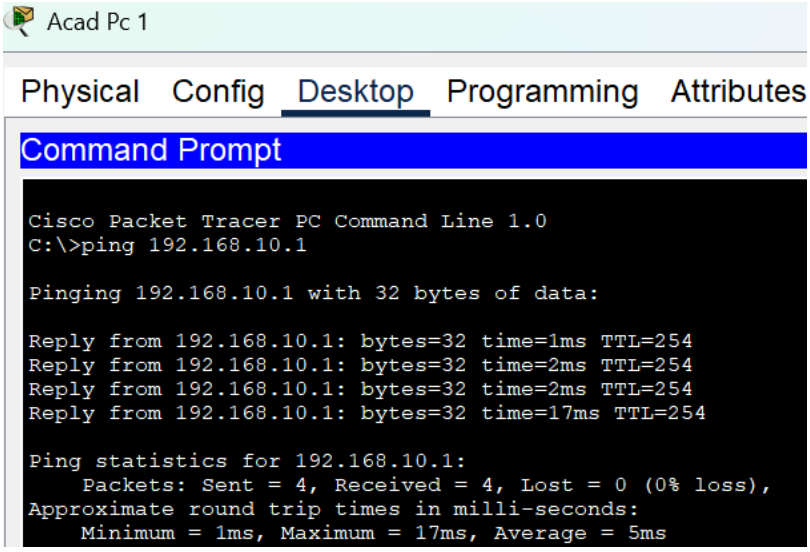


- Successfully permitted ping



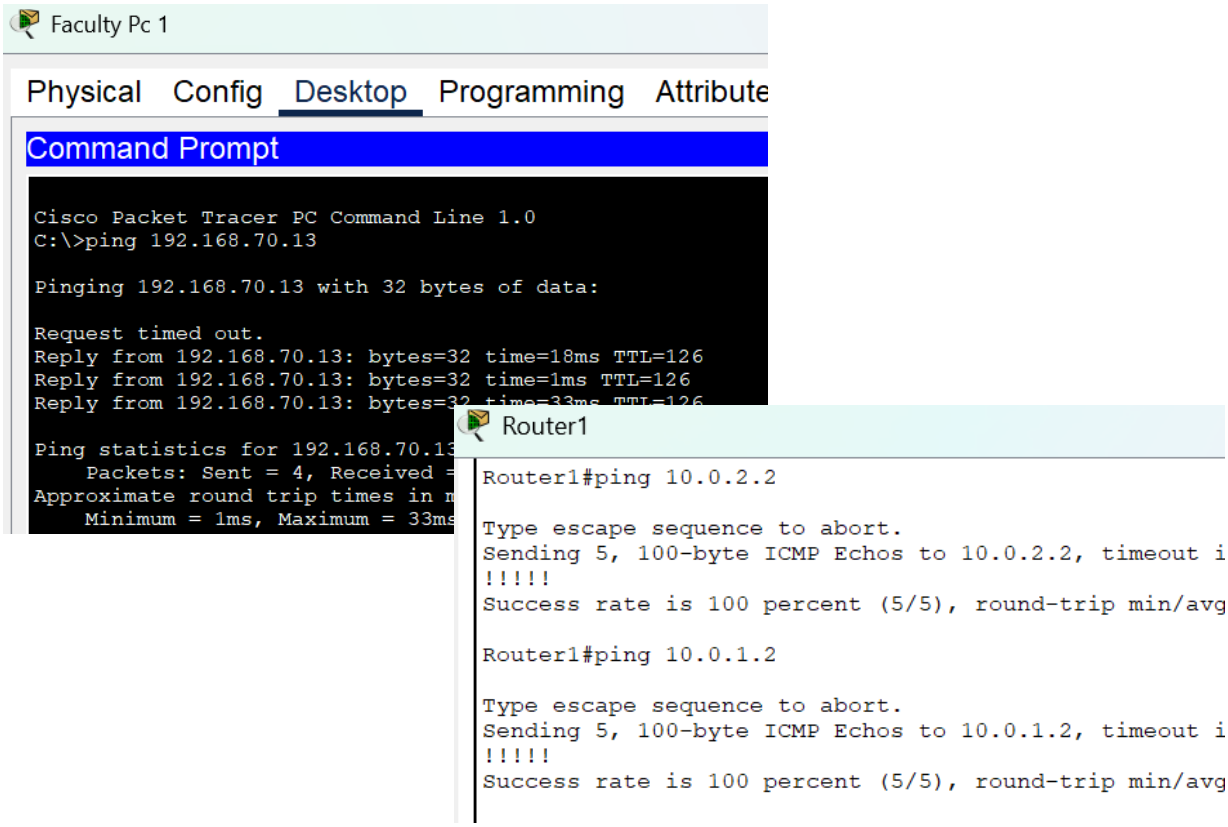
- **WAN Connectivity Test (Inter-Campus Communication)**

This test confirms WAN connectivity between all three campuses. PCs from remote campuses successfully pinged PCs and servers at the Main Campus.



- **End-to-End Network Communication Test**

This test validates complete end-to-end communication across VLANs, campuses, and WAN links. A PC from a remote campus successfully accessed a permitted service hosted at the Main Campus.



Summary

This project successfully achieved the objective of designing, configuring, and verifying a secure and scalable multi-campus network infrastructure using Cisco Packet Tracer. The initial flat network was transformed into a structured hierarchical design that efficiently connects the Main Campus, Academic Block, and Student Services Center through reliable WAN links.

At the Main Campus, VLANs were implemented to logically separate departments such as Administration, Faculty, IT Support, Library, Accounts, and Servers. This significantly reduced broadcast traffic, improved performance, and enhanced security. Inter-VLAN communication was enabled using the Router-on-a-Stick approach, allowing controlled and efficient data flow between departments. A centralized server was deployed to provide essential services such as DNS and Web hosting, making network services accessible across all campuses.

The Academic Block and Student Services Center were designed as remote sites with single VLAN architectures, simplifying management while maintaining full connectivity with the Main Campus. WAN connectivity was established using serial links, and dynamic routing was implemented using OSPF to ensure automatic route discovery and reliable inter-site communication.

Network security was strengthened through the implementation of Access Control Lists (ACLs), which protected sensitive resources such as the Accounts department and the server network while still allowing permitted access from authorized VLANs and remote campuses. Comprehensive network verification confirmed successful LAN communication, inter-VLAN routing, WAN connectivity, and secure access control.

Overall, this project demonstrates the practical application of core networking concepts including VLANs, subnetting, routing protocols, WAN design, server integration, and security policies. The final network design is robust, scalable, secure, and fully aligned with the requirements outlined in the project documentation, fulfilling both academic and real-world networking objectives.

←THE END→